Discrete Mathematics

Test 2

30 minutes

· October 2026

1. Consider the following recurrence relation:

$$a_n - 6a_{n-1} + 9a_{n-2} = 6 \cdot 3^n,$$

with $a_0 = 2$ and $a_1 = 9$.

(a) (2 points) Find the homogeneous solution of this recurrence relation.

(b) (3 points) Propose a general particular solution of the recurrence relation and explain your reasoning.

(c) (5 points) Use your particulation solution ,together with the initial conditions, to find the closed form of a_n.

(d) (3 points) Hence, use the closed form obtained to find the generating function of this recurrence relation. Give your answer as a single rational fraction.

2. Suppose a recurrence relation has characteristic roots 2, 2, 2 and -1.

(a) (1 point) Write down the homogeneous solution, denoted by $a_n^{(h)}$.

(b) (1 point) If the recurrence relation (R.R) is inhomogeneous with right-hand side $7 \cdot (2)^n$, suggest a particular solution $a_n^{(p)}$. You are not required to find the coefficients.